

# Andi Zhang

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## </> Summary

I am currently a Postdoctoral Research Associate at the University of Manchester. I obtained my PhD from the University of Cambridge, where I proposed the **Probabilistic Adversarial Attack**, a principled probabilistic perspective to adversarial attacks. After my PhD, I developed the **Concept-based Adversarial Attack**, which leverages the probabilistic perspective of adversarial attacks to extend traditional single-image adversarial attacks to broader semantic concepts or entities, expanding the applicability of adversarial methods.

## 🏛 Education

### Trinity College, University of Cambridge

#### PhD in Computer Science

Oct. 2019 – Dec. 2024

- Thesis Title: **Anomalous Inputs in Deep Learning: a Probabilistic Perspective**
- Supervisor: Dr. Damon Wischik
- Keywords: Probabilistic Generative Models, Large Language Models, Out-of-distribution Detection, Adversarial Attack

#### MASt in Mathematics (Part III of the Mathematical Tripos)

Oct. 2018 – Jun. 2019

- Relevant modules: Statistical Learning in Practise, Convex Optimization, Astrostatistics, Bayesian Modelling and Computation, Modern Statistical Methods

#### MPhil in Advanced Computer Science

Oct. 2017 – Jun. 2018

- Relevant modules: Machine Learning for Natural Language Processing, Algebraic Path, Deep Learning for Natural Language Processing, Advanced Functional Programming, Probabilistic Machine Learning

### School of Computer Science, University of Manchester

#### BSc (Hons) in Computer Science and Mathematics

Sep. 2014 – Jun. 2017

- Relevant modules: Machine Learning, Quantum Computing, Compilers, Convex Optimization, Probability, Statistical Methods, Functional Analysis, Topology, Number Theory

## 🔗 Experience

### University of Manchester

Manchester, United Kingdom

#### Postdoc Research Associate in Machine Learning | Supervisor: Prof. Samuel Kaski

Nov. 2024 –

- Supported by UKRI Generative AI Hub.
- Concentrating on user modelling and AI safety. Proposed concept-based adversarial attack.

### MediaTek Research

Cambridge, United Kingdom

#### Deep Learning Intern | Host: Dr. Alberto Bernacchia

Aug. 2024 – Nov. 2024

- Participated to a project in accelerating the sampling of diffusion models by proposing a new second-order estimation of the posterior. Work submitted to NeurIPS 2025.

### University College London

London, United Kingdom

#### Visitor | Hosts: Dr. David Barber and Dr. Mingtian Zhang

July. 2023 – Oct. 2023

- Worked on a new image generative model that based on Diffusion models.
- Worked on Amortized LoRA, a more principle and probabilistic way to generate LoRA parameters.

### Kirin AI Group, Huawei

Cambridge, United Kingdom

#### Research Intern (Part-time) | Host: Mr. Arnau Raventos

Nov. 2021 – Oct. 2022

- Conducted a comprehensive evaluation of various traditional and AI-based frame interpolation methods, providing insights for the product department to leverage in their decision-making processes.
- Contributed to an AI global illumination project, successfully implemented various samplers within the Mitsuba 3 framework.

### Noah's Ark Lab, Huawei

Shenzhen / Beijing, China

#### Research Intern | Hosts: Dr. Yitong Sun and Dr. Shifeng Zhang

Sep. 2020 – Sep. 2021

- Co-developed the local autoregressive model, which significantly advanced the state-of-the-art in Out-Of-Distribution (OOD) detection and lossless compression. The empirical evidence supporting this enhancement was recognized and published in NeurIPS 2021.
- Contributed to a computer vision project through researching deep learning uncertainty for active learning applications. The outcomes of this work were recognized and published in BMVC 2021.
- Contributed to an indoor positioning project leveraging WiFi signal strength. I specifically conducted research on Deep Gaussian Processes, leading our team to enhance indoor positioning accuracy. The successful results of this work were published in IPIN 2021.

## University of Cambridge

Cambridge, United Kingdom

### Course Supervisor (Teaching Assistant)

Oct. 2019 - Present

- Supervised IA Algorithms (2020 Lent, 2023 Lent), IA Machine Learning in Real World (2020 Lent), IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas), IB Data Science (2022 Michaelmas, 2023 Michaelmas), IB Further Graphics (2022 Michaelmas) and II Machine Learning and Bayesian Inference (2023 Lent)

## MRC Biostatistics Unit, University of Cambridge

Cambridge, United Kingdom

### Research Assistant | Hosts: Dr. Sofia Villar and Dr. David Robertson

Jul. 2019 - Sep. 2019

- Developed a simulation system to tackle the multi-armed bandit problem (MABP) in clinical trial settings and pioneered a unique test statistic for binary response MABP situations. Although the time constraints prevented this work from being formalized into a publication, it provided substantial practical and theoretical insight.

## Tencent (WeChat AI)

Beijing, China

### R & D Intern | Host: Dr. Jie Zhou

Jul. 2018 - Sep. 2018

- Engaged in comprehensive exploration and research in deep reinforcement learning and evolution strategies. While this research did not culminate in a tangible output, the experience gained in these cutting-edge fields has proven invaluable.

## Amazon (Alexa)

Cambridge, United Kingdom

### Applied Scientist Intern | Host: Dr. Emilio Monti

Jun. 2017 - Sep. 2017

- Employed the Seq2Seq model, the predecessor of the Transformer, with the Attention mechanism for semantic parsing tasks. Semantic parsing involves translating natural language into a structured internal language used for database or knowledge base querying.

## Robotics Institute, Carnegie Mellon University

Pittsburgh, United States

### Student Intern | Host: Dr. Stelian Coros

Jun. 2016 - Sep. 2016

- Contributed to the implementation of a general stance controller for 3D-printable robotic creatures. The controller's adaptability enables it to be used with any human-designed multi-foot robots, hence its designation as 'general'.

## Publications

### Concept-based Adversarial Attack: A Probabilistic Perspective

Arxiv Preprint

Andi Zhang, Xuan Ding, Steven McDonagh, Samuel Kaski

<https://arxiv.org/abs/2507.02965>

### Noise Diffusion for Enhancing Semantic Faithfulness in Text-to-Image Synthesis

CVPR 2025

Boming Miao, Chunxiao Li, Xiaoxiao Wang, Andi Zhang, Rui Sun, ZiZhe Wang, Yao Zhu

In the IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)

### Improving Probabilistic Diffusion Models With Optimal Diagonal Covariance Matching

ICLR 2025 Oral

Zijing Ou\*, Mingtian Zhang\*, Andi Zhang, Tim Z. Xiao, Yingzhen Li, David Barber

In the 13th International Conference on Learning Representations

### Your Finetuned Large Language Model is Already a Powerful OOD Detector

AISTATS 2025

Andi Zhang, Tim Z. Xiao, Weiyang Liu, Robert Bamler, Damon Wischik

In Proceedings of the 28th International Conference on Artificial Intelligence and Statistics

### Constructing Semantics-Aware Adversarial Examples with a Probabilistic Perspective

NeurIPS 2024

Andi Zhang, Mingtian Zhang, Damon Wischik

In the 38th Annual Conference on Neural Information Processing Systems, 2024

### SR-OOD: Out-of-Distribution Detection via Sample Repairing

Arxiv Preprint

Rui Sun\*, Andi Zhang\*, Haiming Zhang\*, Yao Zhu, Ruimao Zhang, Zhen Li

<https://arxiv.org/abs/2305.18228>

### Falsehoods that ML researchers believe about OOD detection

NeurIPS 2022 Workshop

Andi Zhang, Damon Wischik

In ML Safety Workshop, 36th Conference on Neural Information Processing Systems

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\*Equal contribution.

### Out-of-Distribution Detection with Class Ratio Estimation

NeurIPS 2022 Workshop

Mingtian Zhang\*, **Andi Zhang\***, Tim Z. Xiao, Yitong Sun, Steven McDonagh

In ML Safety Workshop, 36th Conference on Neural Information Processing Systems

### On the Out-of-distribution Generalization of Probabilistic Image Modelling

NeurIPS 2021

Mingtian Zhang\*, **Andi Zhang\***, Steven McDonagh

In Neural Information Processing Systems annual meeting, 2021

### Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep

IPIN 2021

Ran Guan, **Andi Zhang**, Mengchao Li, Yongliang Wang

In International Conference on Indoor Positioning and Indoor Navigation, 2021

### Towards dynamic and scalable active learning with neural architecture adaption for object detection

BMVC 2021

Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, **Andi Zhang**, Wei Zhang, Hongtao Lu, Chunjing Xu

In British Machine Vision Conference, 2021

### Solipsistic Reinforcement Learning

ICLR 2021 Workshop

Mingtian Zhang\*, Peter Noel Hayes\*, Tim Z. Xiao, **Andi Zhang**, David Barber

In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations, 2021

## Honors & Awards

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### Top Reviewer (10%) of AISTATS 2023

Feb. 2023

### The Williams/Kilburn Medal for outstanding final year student

Jul. 2017

School of Computer Science, University of Manchester

The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.

### University awards for outstanding academic achievement

Jul. 2017

University of Manchester

The university will consider the top 0.5% of undergraduates (nominated by each school) for this award.

## Academic Service

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I have served as a reviewer for the following conferences:

- NeurIPS (2024, 2025)
- ICML (2025)
- ICLR (2025)
- AISTATS (2023, 2024, 2025, 2026)
- CVPR (2025)
- ICCV (2025)
- ACM MM (2025)
- AAAI (2026)